

OPERATING SYSTEMS

Storage Management

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Access Methods

When a file is used information is read and accessed

Some systems provide only one access methods for files

Access Methods

- Sequential Access

- Direct Access

- Index Sequential access



Sequential Access

Data is accessed one record right after another record in an order.

Read all bytes/records from the beginning

Cannot jump around, could rewind or forward

When we use read command, it move ahead pointer by one using **ReadNext**

When we use write command, it will allocate memory and move the pointer to the end of the file using **WriteNext**

Editors, compilers access the files in this fashion

Such a method is reasonable for tape

Direct Access

Also known as relative access method

Read in any order

Read and write rapidly can happen

The file is viewed as a numbered sequence of blocks

There is no restriction in the order of reading and writing

Mostly used in database systems

Index Sequential Access

Built on top of Sequential Access method

Create index for the file

Index contains pointers to various blocks

Index is searched sequentially and its pointer is used to access the file directly



THANK YOU

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